

PPE Compliance Monitoring System

Complete Setup Guide — From Scratch

This guide is written for **complete beginners** who may have never used Python, VS Code, or the command line before. Follow every step carefully, in order, and you will have the PPE Compliance Monitoring System running on your Windows computer.

Part 1: Installing the Software You Need

Before working with the project code, you need to install two programs on your computer: **Python** and **Visual Studio Code (VS Code)**.

Download and Install Python 3.10

Important: You must install **Python version 3.10** specifically. Newer versions of Python (3.11, 3.12, 3.13, etc.) can cause compatibility problems with some of the libraries this project uses. Even if you already have a newer Python installed, please install 3.10 alongside it.

Step 1: Download the installer

Open your web browser and go to:

<https://www.python.org/downloads/release/python-31011/>

Scroll down to the **Files** section at the bottom of the page. Click on “**Windows installer (64-bit)**” to download it.

Step 2: Run the installer

Double-click the downloaded file to open the installer.

VERY IMPORTANT: On the very first screen of the installer, you will see a checkbox at the bottom that says “**Add Python 3.10 to PATH**”. **CHECK THAT BOX**. This is the most commonly missed step and the app will not work without it.

After checking that box, click “**Install Now**” and wait for it to finish. When it says “Setup was successful,” click **Close**.

Step 3: Verify the installation

Press the **Windows key** on your keyboard, type `cmd`, and press **Enter**. This opens the Command Prompt. Type the following and press Enter:

```
python --version
```

You should see something like `Python 3.10.11`. If you get an error like “`python` is not recognized,” it means the “Add to PATH” checkbox was not checked. Uninstall Python and repeat the steps above, making sure to check that box.

Download and Install Visual Studio Code (VS Code)

VS Code is a free code editor that makes it easy to work with project files and use the built-in terminal.

1. Go to <https://code.visualstudio.com/> and click the big blue “**Download for Windows**” button.

2. Run the downloaded installer.
3. During installation, check all the boxes that mention “**Add to PATH**” and “**Register Code as an editor.**” Then click **Install**.
4. When finished, click **Finish** to launch VS Code.

Part 2: Downloading the Project Code

1. Go to the project's GitHub page: https://github.com/KIMJK7/PPE_compliance
2. Click the green “<> Code” button.
3. In the dropdown, click “Download ZIP.”
4. Once downloaded, go to your **Downloads** folder and find the ZIP file.
5. **Right-click** the ZIP file and choose “**Extract All...**”. Choose a location like your **Desktop** or **Documents** folder, then click **Extract**.
6. You should now have a folder called **PPE_compliance-main** containing files like `app.py`, `requirements.txt`, and folders like `core`, `camera`, `detectors`, etc.

Part 3: Opening the Project in VS Code

1. Open **VS Code**.
2. Click **File** in the top menu bar, then click **Open Folder...**
3. Navigate to the **PPE_compliance-main** folder you extracted. Select it and click **Select Folder**.
4. You should now see all the project files listed in the left sidebar of VS Code.

Opening the VS Code Terminal

You will run all commands in the VS Code terminal (this is like the Command Prompt but built into VS Code).

1. In VS Code, click **Terminal** in the top menu bar, then click **New Terminal**.
2. A terminal panel will appear at the bottom of VS Code.
3. Make sure the terminal shows something like `PS C:\Users\YourName\Desktop\PPE_compliance-main>`. This confirms you are inside the correct project folder.

Part 4: Creating a Virtual Environment and Installing Dependencies

A “virtual environment” is like a clean, separate workspace for this project’s Python packages so they don’t interfere with anything else on your computer.

Step 1: Create the virtual environment

Type this command in the VS Code terminal and press **Enter**:

```
python -m venv venv
```

This creates a folder called `venv` inside your project. It may take 30 seconds — just wait until you see the cursor blinking again.

Step 2: Activate the virtual environment

Type this command and press **Enter**:

```
venv\Scripts\activate
```

After running this, you should see `(venv)` appear at the beginning of the line in your terminal. This means the virtual environment is active. **Every time you open a new terminal to work on this project, you must run this activate command again.**

If you get an error about “execution of scripts is disabled”: Type the following command first, then try the activate command again:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Step 3: Install the required Python packages

Now, with `(venv)` showing in your terminal, run this command:

```
pip install -r requirements.txt
```

This reads the `requirements.txt` file and automatically installs everything the app needs. It may take a few minutes and you will see a lot of text scrolling — this is normal.

Step 4: Run a safety install command (just to be certain)

Even after the above step, run this command as well to make sure the core packages are definitely installed:

```
pip install flask flask-cors python-dotenv opencv-python onnxruntime reportlab
```

You may see messages like “Requirement already satisfied” — that is perfectly fine and means everything is in order.

Part 5: Configuring the Camera (the .env File)

The project uses a file called `.env` to know which camera to connect to.

Step 1: Create the .env file

In the VS Code file explorer (left sidebar), look for a file called `.env.example`. You can either:

- **Right-click** on it, choose **Copy**, then **Paste** and rename the copy to `.env`
- Or run this command in the terminal:

```
copy .env.example .env
```

Step 2: Edit the .env file

Click on the `.env` file in the sidebar to open it. The most important settings are:

- **RTSP_URL** — The address of your CCTV/IP camera. It usually looks like `rtsp://username:password@192.168.1.100:554/stream`. Your camera manufacturer's documentation will tell you what this URL is.
- **CAMERA_INDEX** — If you want to use your laptop's built-in webcam instead, set this to `0`. For an external USB webcam, try `1` or `2`.

Set either `RTSP_URL` (for an IP/CCTV camera) or `CAMERA_INDEX` (for a webcam), depending on what you are using. **Save** the file (**Ctrl + S**).

Step 3: Place your ONNX model

Make sure your compliance detection model file (usually called `best.onnx`) is placed inside the **models** folder in the project. If there is no `models` folder, create one. The path should be `models\best.onnx`.

Part 6: Running the Application

With everything set up, make sure your terminal shows `(venv)` (meaning the virtual environment is active), and run:

```
python app.py
```

You will see some text appear in the terminal. Look for a line that says something like:

```
* Running on http://127.0.0.1:5000
```

Open the App in Your Web Browser

Open your web browser (Chrome, Edge, Firefox, etc.) and go to any of these addresses:

Page	URL
Home Page	http://127.0.0.1:5000/
Live Camera Feed	http://127.0.0.1:5000/live
Dashboard	http://127.0.0.1:5000/dashboard

The app is now running. You should see the live camera feed with PPE detection overlays.

How to Stop the App

To stop the app, go back to the VS Code terminal and press **Ctrl + C**. This shuts down the server.

Part 7: Switching to a Different Camera

If you want to change which camera the app uses (for example, switching from one IP camera to another), you do **not** need to reinstall anything. Just follow these steps:

1. **Stop the app** if it is currently running (press **Ctrl + C** in the terminal).
2. Open the `.env` file in VS Code.
3. Change the **RTSP_URL** value to the new camera's RTSP address.
4. **Save** the file (**Ctrl + S**).
5. Run the app again: `python app.py`
6. Open **`http://127.0.0.1:5000/live`** in your browser and you will now see the new camera's feed.

Example of what to change in the `.env` file:

```
Old:  RTSP_URL=rtsp://admin:pass123@192.168.1.100:554/stream
New:  RTSP_URL=rtsp://admin:pass123@192.168.1.101:554/stream
```


Part 8: Running a Second Camera (Setting Up a Second Copy)

Very Important Limitation: This application runs on `http://127.0.0.1:5000`. You **cannot** run two copies of this app at the same time because they would both try to use the same address (port 5000) and one of them will fail. **You can only monitor one camera at a time.** To switch cameras, follow Part 7 above.

If you still want to have a **second copy of the project pre-configured** for a different camera (so you can quickly switch without editing the `.env` file each time), here is how:

Create a second copy of the project

1. Go back to the **original ZIP file** you downloaded in Part 2.
2. **Extract it again** to a different location. For example, extract it into a folder like `Desktop\PPE_Camera2\`.
3. Rename the extracted folder to something descriptive, like **PPE_compliance_Camera2**.

Set up the second copy

1. Open a **new VS Code window** (File → New Window) and open the second folder (File → Open Folder).
2. Open the terminal in this new VS Code window (Terminal → New Terminal).
3. Create and activate a virtual environment, then install dependencies:

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
pip install flask flask-cors python-dotenv opencv-python onnxruntime reportlab
```

4. Create the `.env` file and set **RTSP_URL** to your **second camera's** address:

```
copy .env.example .env
```

Then edit `.env` and set:

```
RTSP_URL=rtsp://admin:pass123@192.168.1.102:554/stream
```

Switching between cameras with two copies

With two pre-configured copies, switching cameras is simple:

- **Stop** whichever copy is currently running (Ctrl + C).
- Go to the **other** copy's VS Code window.
- Make sure the virtual environment is active (`venv\Scripts\activate`).
- Run `python app.py` and refresh your browser.

Troubleshooting

“python is not recognized as an internal or external command”

Python was not added to your system PATH. Uninstall Python, reinstall it, and make sure to check the “Add Python 3.10 to PATH” checkbox on the first screen of the installer.

“Execution of scripts is disabled on this system”

Run this command in your terminal and then try activating the virtual environment again:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Webcam not showing / black screen

Try changing CAMERA_INDEX in the .env file to 0, 1, or 2. Also close any other apps that might be using your camera (Zoom, Teams, browser camera access, etc.).

RTSP stream is laggy or keeps freezing

In your .env file, try lowering MAX_FPS to a smaller number (like 10) and reducing FRAME_WIDTH and FRAME_HEIGHT to smaller values (like 640 and 480).

“ModuleNotFoundError: No module named ‘flask’”

Your virtual environment is probably not activated. Make sure you see (venv) at the beginning of your terminal line. If not, run venv\Scripts\activate and try again.

Port 5000 is already in use

Another application (or another copy of this app) is already using port 5000. Stop it first (Ctrl + C), or check your system for other programs using that port.

Quick Reference

Useful URLs When the App Is Running

Page	URL
Home	http://127.0.0.1:5000/
Live Camera Feed	http://127.0.0.1:5000/live
Dashboard	http://127.0.0.1:5000/dashboard

Common Commands

What You Want To Do	Command
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Activate virtual environment	venv\Scripts\activate
Install dependencies	pip install -r requirements.txt
Run the app	python app.py
Stop the app	Press Ctrl + C in the terminal